

Chloe Kim

Resume – Academic Emphasis

📍 Cranbrook Kingswood Upper School, 39221 Woodward Ave. Bloomfield Hills, MI

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Academic Interests & Selected Outputs

Focus: AI × Human Decision-making under Uncertainty & Fairness

- Cognition & intelligence — perception, reasoning, risk
- Applied domains: social applications of AI, AI insurance, legal networks, healthcare triage

Keywords: Computational cognitive science; reinforcement learning & Bayesian inference; NLP; causal inference

Selected Outputs

- *Professional Academic Presentations:* EURO 2025; Helmholtz AI 2025 (*Spotlight*)
— Human *vs.* model-based reasoning; recognition & uncertainty, fairness; AI applications
- *Patent:* AI insurance framework linking reliability risks to actuarial metrics (Sole Inventor)
- *Software Reg. (4 works):* Network dynamics, power-law & inequality study (Sole Developer)

Education

Cranbrook Kingswood Upper School *High school student*

*Aug 2023–
Bloomfield Hills, MI*

Humanities, Biology, Linguistics, AI, Comparison of learning algorithms

Harvard University *Undergraduate-level studies*

*July 2024–Aug 2024
Cambridge, MA*

Molecular Biology (BIOS-S1a) (A-) and Historical Linguistics (LING-120) (A)
Studies at Harvard Summer School (for credit)

南京时代双语学校 (Nanjing Times College) *Middle school student*

*Aug 2021–Jun 2023
南京(Nanjing), China*

Kyunghee School *Elementary school student*

*Mar 2015–Feb 2021
Seoul, Republic of Korea*

Selected Research & Intellectual Property

Patent *Sole Inventor*

“System and method for evaluating AI output consistency and providing dynamic insurance”
#10-2025-0104581, patent pending (Korea)

- Quantifies AI risks (factuality, logical/semantic consistency, bias) into actuarial metrics.
- Designs dynamic premium pricing & claim validation linked to AI performance.
- *AI Insurance* – recompensation for AI hallucination

Software Registrations *Sole Developer*

“Interpreting Social Intent and Behavior: Temporal Dynamics of Network Heatmaps” (#035490)

- Analyzes temporal networks, creating centrality- & time-based maps of social relations.
“Analysis of Social Intentionality: Visualization of Case-law Networks” (#035307)
- Processes legal datasets, computing centrality & visualizing case-law structures.
“Simulation of Behavioral Relationships in Power-law Dynamics using AI” (#035489)
- Models behavior relations under power-law dynamics with AI sampling & visualization.
“Inequalities in Human Behavior: A Graph-based and AI Interpretation” (#034662)
- Analyzes behavioral inequality using AI/graphs, quantifying disparities via references.

Research Presented *First author & Presenter*

“Dynamic Network Analysis of Legal Citation Networks (especially medical)”

34th European Conference on Operational Research (EURO 2025), Leeds, UK

- Analyzed 86,000+ Supreme Court cases using network and power-law analysis, showing concentrated reliance on landmark precedents.
- Illustrate how structural citation patterns shape legal & medical decision-making.

“Unveiling Behavioral Patterns in Legal Citations through Complex Network and AI Analyses”

Manuscript prepared for journal submission (mimeo), Oct 2025

“An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI”

Helmholtz AI Conference 2025, Karlsruhe, Germany

- Compared 40+ multilingual participants with GPT-4 on sound-to-shape tasks, revealing humans’ intuitive mappings *vs.* AI’s semantic reasoning.
- Findings support dual-mode cognition, highlighting differences in human and machine decision-making.

“Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction” (*Spotlight*)

Helmholtz AI Conference 2025, Karlsruhe, Germany

- Trained on 500+ patient cases with fused imaging (Vision Transformer) and clinical features (XGBoost), achieving 92.7% accuracy and AUROC 0.90.
- Results demonstrate scalable, interpretable AI for airway management, highlighting contrasts between human judgment and standardized decision-making.

Work Experience

Lawfirm Hoam *Research Assistant with Lawyer*

*Summer 2024–
Seoul, Rep. of Korea & Remote*

- Perform data analytics and lawsuit cases research with AI and quant. network analyses.
- Presented results in an academic conference.

Quilt.ai *Research Intern*

*Jul 2024–Dec 2024
Boston, MA & Remote*

- Evaluated and researched AI narrative generation for fashion ads. that support human perception and cognition
- Ran market surveys; produced insight memos used by the design/product teams.

Media art Yeseung Lee's team *Assistant Editor/Supporter**Dec 2024–
Seoul, Rep. of Korea & Remote*

- Supported a project on art and human cognition.
- Edited artist statements and exhibition texts (Singapore, Seoul Dongdaemun Design Plaza (DDP), National Museum of Modern Art (MMCA) exhibitions)

Elroi AI *Research Intern**May 2025~
Troy, MI & Remote*

- Prototyped and studied modules for generation and HAI workflows; conducted user tests
- Delivered a technical brief.

SAT Mathematics Tutoring *Tutor**June 2025~September 2025
NYC, NY & Remote*

- Tutored high school students in SAT Math preparation and strategies

Computer Science Projects *Independent researcher**Jun 2022–
Bloomfields Hills, MI & remote*

- Built coding projects including BMI Calculator, Quiz Game, Server–Client Trading Apps, and AI&ML-based game simulations
- 9th – 12th grade

Leadership & Service

Varsity Rowing Team *Coxswain*

- Coach/manage boats (steering/making calls & corrections), mentor novice rowers, delegate roles, manage race day logistics & encourage team support

Linguistics Club *Founder, Co-President*

- Founded Linguistics Club to promote different branches of linguistics; raise awareness of diverse languages; NACLO prep; AI+ Linguistics research

Ethics Bowl Club *President/Founder*

- Founded club; recruited, hosted state-wide practice events, organized spaces for ethical thinking & dialogue in & out of school & community

Crane Clarion (School newspaper) *Assoc. Arts Editor, Graphic Designer, Staff Writer*

- Led artists & writers; mentored junior writers; established journalistic protocols; publicized arts & school spirit; wrote cultural sens. guidelines

LITE (Leaders in the Environment) *Food Board Captain*

- Liaison to school meal providers; promoted inclusive meals; systemized negotiation process btw. student body, faculty, and providers; org 10+ events

Cranbrook Kingswood Chamber Orchestra, Thornlea Quartet *Violinist; Organizer*

- Violinist in school orchestra; performed at school/community; organized fundraising concerts for Brilliant Detroit; mentored peers

Tutoring *Comm. Service – Volunteer tutor*

- Independent & Volunteer – Local & Overseas; hybrid—in-person/online via Zoom/Teams)

- Tutored 5-7 under-resourced students: English (Chinese & Korean) + SAT Math (US); ran Zoom lessons; built practice sets; measurable score gains

Gold Key *Sophomore Lead*

“Ergasterion” – Theater & Drama club *Stage Crew & Member*

Student Council *Member*

World Scholars Cup *Team leader*

- Led the debate team; won in conferences

Awards and Honors

National Latin Exam Cum Laude and 2x Magna Cum Laude – *G9, 10, 11; National*

Ethics Bowl Central Division 4th place – *G10, 11; State*

3rd in State for National Economics Challenge – *G10; State*

District MSBOA 1st Division (solo) & 1st/2nd Division (quartet) – *G9, 10, 11; State*

NEC – National Economics Challenge; 3rd (D. Ricardo) – *G10; State*

Strickland Writing Award – *G9; School*

Dean’s List & Honors – *G9, 10, 11; School*

AP Scholar with Distinction Award – *G11; School*

World Scholars Cup: Qualified for the global round – *2020-21; National (China)*

External Studies

Human Asia Academy

*Jun 2023,
Seoul, Rep. of Korea*

- “Human rights” for North Korean people

MIT OpenCourseWare – *Self Studies*

- Python & Julia programming language; concepts, grammar, applications
- Abstraction, object-oriented programming, recursion, debugging, interface to C

FastCampus Korea – *Machine Learning Courses*

- AI, Machine learning and Data Analytics (Python, Tensorflow, Dockers, LLM-agents)

Skills

AI and Data Analytics

- Statistics & Data analytics
- Network and complex system analyses

Technical Computer Science *Java, Python, Julia*

- Applications to Semantics & philosophy

Additional

Citizenship: United States

Languages: English (native); Korean (native/bilingual); Chinese (Mandarin) (native); French (intermediate); Japanese (intermediate); Latin (NLE Latin III)